



Republic of Rwanda
Ministry of Education



CEILING INSTALLATION

BDCCI501

Install Ceiling

Competence

RQF Level: 5

Learning Hours



Credits: 6

Sector: Construction and Building Services

Trade: Building construction

Module Type: Specific

Curriculum: CBSBDC5001-TVET CERTIFICATE V IN BUILDING CONSTRUCTION

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Purpose statement	This specific module describes the skills, knowledge and attitudes required to construct ceiling. It is meant for learners who have successfully completed level IV in building construction and pursuing level V in building construction. At the end of this module students will be able to Perform ceiling preliminary works, set out ceiling, construct ceiling frame and apply ceiling coverings with minimum supervision.					
Leaning assumed to be in place	scaffolding operations, Roof construction, ARCHICAD software					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
Perform ceiling preliminary works	1.1 Ceiling drawings are properly interpreted according to the provided model.
	1.2 Materials, tools and equipment are properly prepared based on ceiling type and its details.
	1.3 Safety precautions are effectively applied according to the working condition.
Set out ceiling frame	2.1 Reference line is correctly set according to construction drawing details
	2.2 Height of the ceiling is correctly fixed based on type of ceiling
	2.3 Level of ceiling and frame are precisely marked according to the type of ceiling
	2.4 Level of frame ceiling is properly fixed according to the type of ceiling

Construct ceiling frame	3.1 Scaffolding structure is properly erected according to the work to be performed
	3.2 Ceiling frame layouts is properly identified based on ceiling drawing
	3.3 Ceiling frame members are properly assembled based on the frame layouts
Install ceiling covering materials	4.1. Ceiling panels are properly cut based on cutting lists
	4.2. Ceiling panels are rigidly fixed to the frame based on ceiling drawing
	4.3. Finishing works are properly performed in accordance with design concept

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify types of ceiling ✓ Describe ceiling design principles ✓ Identify materials, tools and equipment used in ceiling ✓ Describe ceiling construction techniques ✓ Identify ceiling finishes ✓ Identify design tools ✓ Describe ceiling features ✓ Describe safety measures ✓ Describe project reports	✓ Perform ceiling cutting methods ✓ Interpret ceiling drawings ✓ Prepare ceiling boards panels ✓ Perform ceiling assembling techniques ✓ Fix ceiling members ✓ Carry out ceiling finishing activities ✓ Install the ceiling frame ✓ Apply ceiling construction procedures	✓ Use materials efficiently and effectively ✓ Make right decision within project design and execution ✓ Work under unexpected timeframe and conditions ✓ Pay attention to ceiling design project details ✓ Work collaboratively with team ✓ Use resources effectively during ceiling construction

	✓ Prepare ceiling project reports	✓ Use creativity and innovation throughout ceiling creation ✓ Meet set ceiling project deadlines ✓ Focus on ceiling project perfection
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Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform ceiling preliminary works 2. Set out ceiling frame 3. Construct Ceiling frame 4. Apply ceiling covering
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Learning outcome 1: Perform ceiling preliminary works

Learning hours: 5 hours

Indicative content

- **Interpretation of ceiling drawings**
 - ✓ Drawing dimensions
 - ✓ Drawing symbols and signs
 - ✓ Drawing scale
 - ✓ Drawing plans
 - ✓ Drawing sections and views
- **Preparation of materials, tools and equipment**
 - ✓ Identification of ceiling materials
 - ✓ Identification of tools used in ceiling
 - ✓ Identification of equipment used in ceiling
 - ✓ Selection of materials, tools and equipment
- **Applying safety precautions in working area**
 - ✓ Use Personal Protective Equipment
 - ✓ Working area safety
 - ✚ Sign posts
 - ✚ Hazards and risks management

✓ Cleaning the workplace

Resources required for the indicative content

Equipment	PPEs ,Circular saw ,Ladder/scaffolding, Air compressor, Band saw, Wheel barrow ,Surface planer ,Thicknesses planner, Belt sander, Grinder machine, Drilling machine, Nail gun, welding machine.
Materials	Stucco, Gypsum compound, Paint, Wooden board , Acoustic tile, Gypsum, Plywood, PVC Panel, MDF, Nails, Warning tapes, Glue, Sanding papers, Screws , Bolt, Hinges , Locks, Varnish , Lacquer , Thinner, Galvanized wire, Water , PVC pipes, Scratching materials, cutting discs, electrodes, protective materials, fuels.
Tools	Hammers, Hose pip, Building line, Carpenter's bags, Measuring tape, Spirit level, Hand saws , Machete , Pencil/pen/mark pen, Folding rulers, Screw drivers, Spanners ,Hand planers , Squares , Brush , Chisels , Divider, Files, Rasp, Utility knife, Marking gauge , Painting roller
Facilitation techniques	Group discussion, individual activity - The trainer prepares a ceiling plan and asks students to discuss key elements to review when interpreting the plan. Afterward, the trainer asks them to interpret the ceiling plan based on their observations - The trainer gives a brief overview of the materials, tools, and equipment needed for ceiling installation, and involves trainees in selecting these items individually. - Trainer asks students site visit in the workshop or site for demonstrate the required functions of each tools and equipment used in ceiling installation and trainees prepare site visit report findings. - Trainer prepare the scenario about the activity conducted in ceiling construction, asks trainees to identify potential hazards and recommended PPE based on that activity.
Formative assessment methods	- Written assessment - Performance assessment - Oral assessment

Learning outcome 2: Set out ceiling frame		Learning hours: 15 hours
Indicative content		
• Setting the reference line ✓ Description Methods of fixing reference line of ceiling ✓ Fixing reference line for ceiling ✓ Marking the reference line • Fixing the ceiling height ✓ Description of ceiling types ✓ Identify Standard height of ceiling ✓ Marking the height of ceiling • Levelling the ceiling frames ✓ Determine the level of ceiling frame ✓ Mark ceiling frame ✓ Alignment of ceiling frame ✓ Fix ceiling frame level		
Resources required for the indicative content		
Equipment	• PPEs, Ladder/scaffolding, Air compressor, Band saw, Drilling machine, Nail gun, laser meter.	
Materials	• Marker pen, chalk, nails, screws, welding rod	
Tools	• Hammers, Hose pip, Building line, Carpenter's bags, Measuring tape, Spirit level, Pencil/pen/mark pen, Folding rulers, Screw drivers, Marking gauge, Painting roller	
Facilitation techniques	• scenario, individual activity, group discussion • The trainer shows a video demonstrating the process of setting out a ceiling frame and divides the students into manageable groups, prompting them to discuss the steps involved in this process • Trainer demonstrate to the students on how to set out the reference line and trainer asks students to set reference line themselves. • Trainer demonstrate to the students to fix out ceiling height and trainer asks trainees to fix ceiling height themselves. • Trainer demonstrate to the students on how fix out ceiling height and trainer asks trainees to fix ceiling height themselves. • Trainer and trainees perform activity of levelling ceiling frame and trainer asks trainees to level ceiling frame themselves • Trainer provide students site visit ongoing project.	
Formative assessment methods	• Written assessment • Performance assessment • Oral assessment	

Learning outcome 3: Construct Ceiling frame		Learning hours: 20 hours
Indicative content		
• Erect scaffolding ✓ Select scaffolding ✓ Protect the scaffolding lower part (base plate) ✓ Fix scaffolding parts • Identification of ceiling frame layout ✓ Parts of ceiling frame  Runner  Hanger  Tie beam  Struts ✓ Determine direction of ceiling joists ✓ Marking ceiling shape • Assembling wooden ceiling frame ✓ Fixing the perimeter wall joists ✓ Marking on perimeter wall joists ✓ Fixing the main runners ✓ Fixing cross tee ✓ Levelling the frame • Assembling metal /aluminium ceiling frame ✓ Fixing the perimeter wall joists ✓ Marking on perimeter wall joists ✓ Fixing the main runners ✓ Fixing the cross tee ✓ Levelling the frame Alignment with building lines		
Resources required for the indicative content		
Equipment	• Screw gun or drill, safety equipment (PPE),welding machine	
Materials	• Chalk, charcoal, marker , pen/ ink pen, nails, screws , welding rod	
Tools	• Hammer, tape measure, laser or water level, screwdriver, T-Square	
Facilitation techniques	Group discussion, scenario, simulation and individual activity • Trainer displays a video on erecting scaffolding. In manageable groups the students. He/she asks to discuss on types, elements, characteristics of good scaffolding and scaffolding safety rules	

	• Trainer provide layout and create scenario of ceiling frame. Individually prepare frame layout procedures/steps and make ceiling frame. • Trainer prepare the field visit to the ongoing project site
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 4: Install ceiling covering material	Learning hours: 20 hours
Indicative content	
• Cutting the ceiling panels ✓ Measuring and marking ✓ Cutting the panels ✓ Testing the fit • Fixing the ceiling panels ✓ Fixing method based on ceiling covers and frame materials ✚ Nails ✚ Bolts ✚ pins ✓ Install suspended tile ceiling ✚ Locate ceiling joists or support ✚ Install wall angles ✚ Hang main runners ✚ Insert cross Tees ✚ Cut tiles for edge and obstacles ✚ Install ceiling tile ✚ Final checks ✓ Fixing panels on frame ✓ Fixing joint cover • Perform finishing works ✓ Apply finishing on plywood and wooden plank ceiling ✚ Varnishing	

- ✚ Painting
- ✚ Staining
- ✓ Apply finishing on gypsum ceiling
 - ✚ Plastering
 - ✚ Molding and skirting
 - ✚ Texturing
 - ✚ Painting

- **Prepare the workplace**

- ✓ Removal of waste materials
- ✓ Materials for cleaning workplace
- ✓ Apply cleaning methods
 - ✚ By Water
 - ✚ By Brush
 - ✚ By Air compressor

Resources required for the indicative content

Equipment	● Paint sprayer, Compressor, PPE, Drilling machine, Screw gun ,Jig saw, Router machine
Materials	● Thinner, Masking tape, varnish, paint, Sandpaper, Wood, Screws, Nails, Acoustic ceiling panels, Plywood, wall angles, cross tees, toile ceiling, hanging wires, suspended tile panel.
Tools	● bucket, Tape measure, Drawing set, Spirit levels, Notebook, Screw driver, Hand saw, Plumb bob, Hammer, Try-square, Building line, knife cutter.
Facilitation techniques	Simulation and individual activity ● Trainer shows step by step how to cut and fix ceiling covering using actual materials and tools. This can be done in person or through video demonstration. He/she asks individually to cut and fix ceiling covers. And provide detailed instruction and guidance as students work through the cutting ● Trainer demonstrates the ceiling finishing through video or done in person and asks individually to apply finishing. He/she provide detailed instruction and guidance as students work through fixing process. ● Trainer allows students to practice installing ceilings covering and make finishing themselves under supervision.
Formative assessment methods	● Oral assessment ● Written assessment ● Performance assessment

Integrated/Summative assessment (For specific module)

Integrated situation

MP Hotels located in Nyanza District wants to install a ceiling in their 8x10x3m suite rooms. The company would like to hire ceiling installer to perform ceiling activities for the suite room to improve the room that will be warm, welcoming and reflect the charm of Nyanza District. As a ceiling installer, you are requested to perform the following tasks:

1. Set out ceiling frame
2. Install ceiling frame, from longitudinal to transversal sides of room.
3. Apply ceiling covering materials
4. Ceiling plan, materials, tools and equipment are available on site.
5. Activities will be performed in 8 hours.

Resources

Tools	bucket, Tape measure, Drawing set, Spirit levels, Notebook, Screw driver, Hand saw, Plumb bob, Hammer, Try-square, Building line
Equipment	Paint sprayer, air Compressor, PPE, Drilling machine, Screw gun, Jig saw, Router machine, miter saw, thicknesser machine
Materials/ Consumables	Thinner, Masking tape, varnish, paint, Sandpaper, Wood, Screws, Nails, Acoustic ceiling panels, Plywood

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Perform ceiling preliminary works	1.1. Construction drawings are properly interpreted according to the provided model.	Ceiling dimensions are interpreted			3
		Ceiling symbols and sign are interpreted			3
	1.2. Materials, tools and equipment are well selected based on ceiling type and its details.	Materials are selected			3
		Tools are selected			3
		Equipment are selected			3
	1.4. Safety precautions are effectively applied according to the working condition.	PPE are worn			3
		Sign posts are installed			3

Learning outcome 2: Set out ceiling frame	2.1. Reference line is correctly set according to construction drawing details	Reference line is set			3
	2.2. Height of the ceiling is correctly fixed based on type of ceiling	Height of ceiling is fixed			3
	2.3. Level of ceiling and frame are precisely marked according to the type of ceiling	Ceiling and frames level is marked			3
	2.4. Level of frame ceiling is properly fixed according to the type of ceiling	Ceiling frame level is fixed			4
Learning outcome 3: Construct Ceiling frame	3.1. Scaffolding structure is properly erected according to the work to be performed	Scaffolding elements are assembled			4
		Scaffolding is stable			4
		Scaffolding height is respectively			4
	3.2. Frame layouts is properly fixed on the design	Marking is done			3
		Building line are set			3
		Direction of joists are identified			4
		Direction of main runners are identified			4
	3.3. Ceiling frame members are properly assembled based on the frame layouts	Perimeter wall joists are fixed			4
		Main runners are fixed			4
		Cross tee are fixed			4
		Frame components are arranged			4
		Frame is levelled			4
Learning outcome 4: Apply ceiling covering material	4.1. Cut ceiling panels based on cutting lists	Gypsum is marked			2
		Gypsum is cutted			2
		Gypsum is tested to fit in the frame			2
		Fixing method is chosen			3

	4.2. Ceiling panels are rigidly fixed to the frame based on design	Gypsum and frame are fitted			3
		Joint covers are fixed			3
	4.3. Finishing works are properly performed in accordance with design concept	Preparation the surface is done			3
		Joint compound is applied/mud			3
		Nails/screws holes are covered			3
		Surface is sanded			3
		Finishes materials are prepare			3
		Finishes are applied			3
		Workplace is cleaned			3
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

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